

Foundations of Artificial Intelligence

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Spring Term 2026

Exercise Sheet 10 — Solutions

Exercise 10.1 (0.5+0.5+1+1+1+1 marks)

Consider the STRIPS planning task $\Pi = \langle V, I, G, A \rangle$ with $V = \{a, b, c, d, e, f\}$, $I = \{a, b\}$, $G = \{e, f\}$, and $A = \{a_1, a_2, a_3\}$ with $\text{cost}(a) = 1$ for all $a \in A$ and

$$\begin{array}{lll} \text{pre}(a_1) = \{a\} & \text{add}(a_1) = \{c\} & \text{del}(a_1) = \{e\} \\ \text{pre}(a_2) = \{b\} & \text{add}(a_2) = \{d\} & \text{del}(a_2) = \{c\} \\ \text{pre}(a_3) = \{c, d\} & \text{add}(a_3) = \{e, f\} & \text{del}(a_3) = \{\}. \end{array}$$

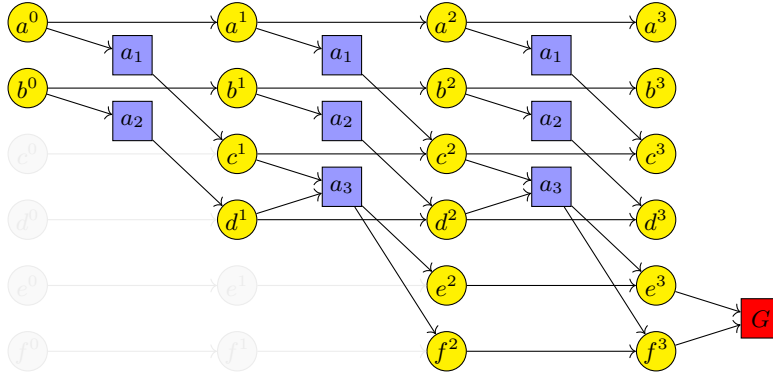
- Provide $h^*(I)$, the cost of an optimal plan.
- Provide $h^+(I)$, the cost of an optimal relaxed plan.
- Provide the relaxed planning graph (RPG) for Π with enough layers for it to stabilize (i.e., up to a layer n such that $V_{n-1} = V_n$).
- Compute $h^{\max}(I)$. Provide the values for all nodes in the RPG.
- Compute $h^{\text{add}}(I)$. Provide the values for all nodes in the RPG.
- Compute $h^{\text{FF}}(I)$. Provide the marked RPG.

Solution:

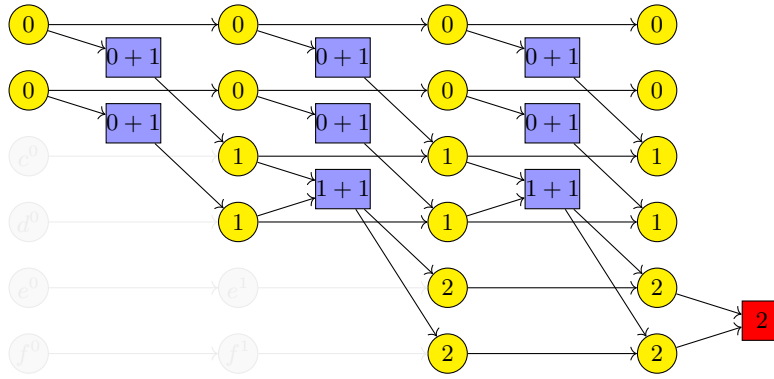
(a) $h^*(I) = 3$

(b) $h^+(I) = 3$

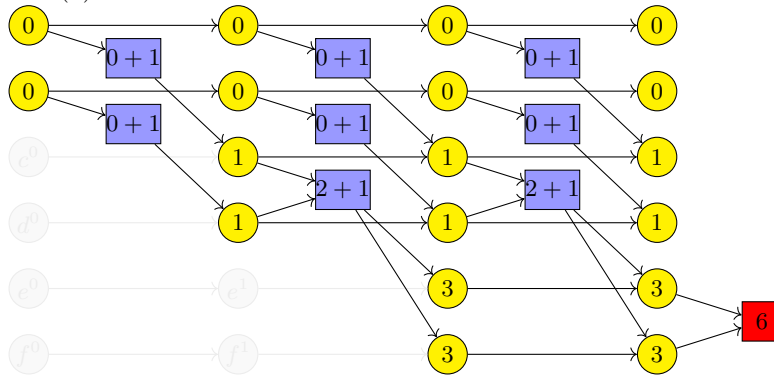
(c)



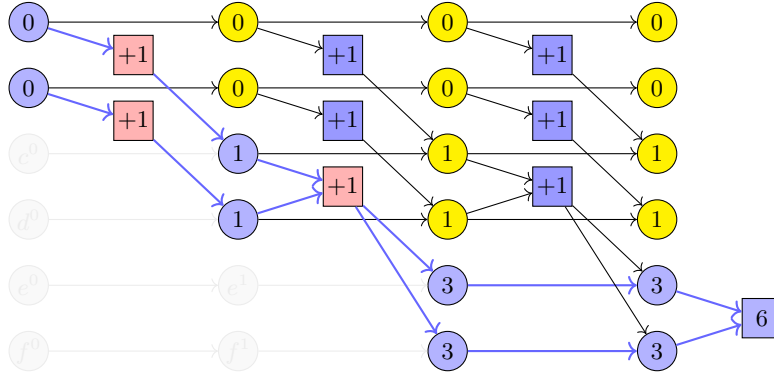
(d) $h^{\max}(I) = 2$



(e) $h^{add}(I) = 6$

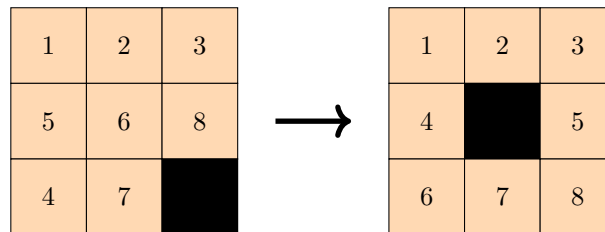


(f) $h^{FF}(I) = 3$



Exercise 10.2 (2 marks)

Formalize the following 8-puzzle instance in SAS⁺:



Solution:

We choose a representation where each grid position is represented by a variable $p_{i,j}$, with $p_{0,0}$ representing the lower left corner of the grid. The value of each variable is the number of the

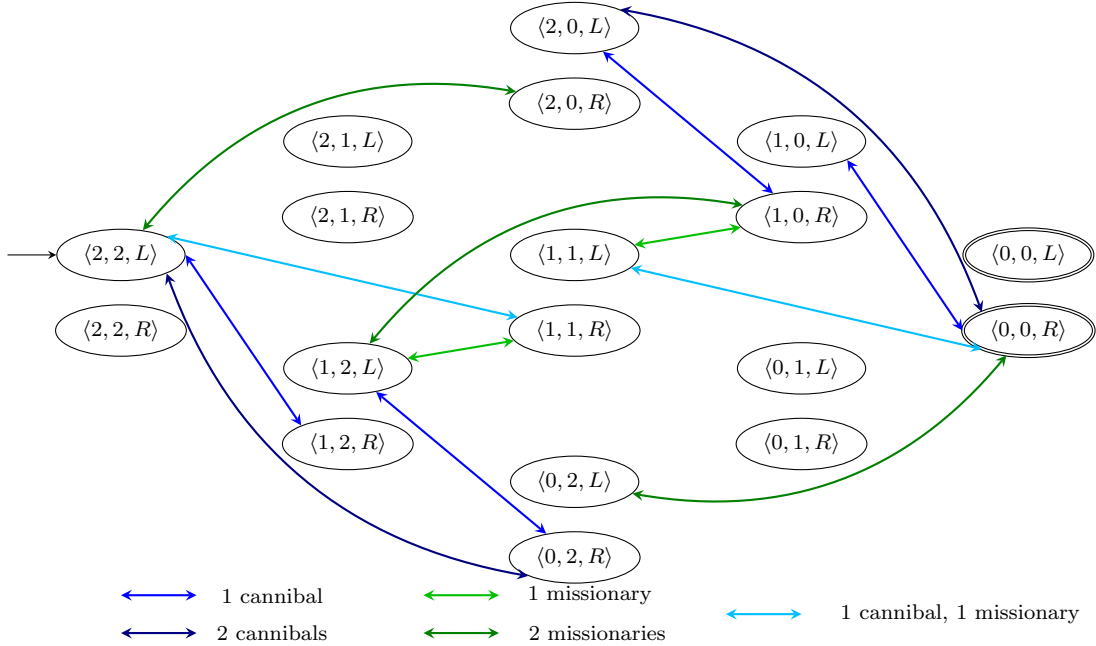
tile currently in this grid position, or b if the position is currently blank. Formally, we define $\Pi = \langle V, \text{dom}, I, G, A \rangle$ with

- $V = \{p_{i,j} \mid 0 \leq i \leq 2, 0 \leq j \leq 2\}$
- $\text{dom}(p_{i,j}) = \{1, \dots, 8, b\}$ for all $t_{i,j} \in V$
- $I = \{p_{0,0} \mapsto 4, p_{1,0} \mapsto 7, p_{2,0} \mapsto b, p_{0,1} \mapsto 5, p_{1,1} \mapsto 6, p_{2,1} \mapsto 8, p_{0,2} \mapsto 1, p_{1,2} \mapsto 2, p_{2,2} \mapsto 3\}$
- $G = \{p_{0,0} \mapsto 6, p_{1,0} \mapsto 7, p_{2,0} \mapsto 8, p_{0,1} \mapsto 4, p_{1,1} \mapsto b, p_{2,1} \mapsto 5, p_{0,2} \mapsto 1, p_{1,2} \mapsto 2, p_{2,2} \mapsto 3\}$
- $A = \{\text{move-up}_{t,i,j} \mid 1 \leq t \leq 8, 0 \leq i \leq 2, 0 \leq j \leq 1\}$
 $\cup \{\text{move-down}_{t,i,j} \mid 1 \leq t \leq 8, 0 \leq i \leq 2, 1 \leq j \leq 2\}$
 $\cup \{\text{move-left}_{t,i,j} \mid 1 \leq t \leq 8, 1 \leq i \leq 2, 0 \leq j \leq 2\}$
 $\cup \{\text{move-right}_{t,i,j} \mid 1 \leq t \leq 8, 0 \leq i \leq 1, 0 \leq j \leq 2\}$ where

$$\begin{aligned} \text{pre}(\text{move-up}_{t,i,j}) &= \{p_{i,j} \mapsto t, p_{i,j+1} \mapsto b\} & \text{eff}(\text{move-up}_{t,i,j}) &= \{p_{i,j} \mapsto b, p_{i,j+1} \mapsto t\} \\ \text{pre}(\text{move-down}_{t,i,j}) &= \{p_{i,j} \mapsto t, p_{i,j-1} \mapsto b\} & \text{eff}(\text{move-down}_{t,i,j}) &= \{p_{i,j} \mapsto b, p_{i,j-1} \mapsto t\} \\ \text{pre}(\text{move-left}_{t,i,j}) &= \{p_{i,j} \mapsto t, p_{i-1,j} \mapsto b\} & \text{eff}(\text{move-left}_{t,i,j}) &= \{p_{i,j} \mapsto b, p_{i-1,j} \mapsto t\} \\ \text{pre}(\text{move-right}_{t,i,j}) &= \{p_{i,j} \mapsto t, p_{i+1,j} \mapsto b\} & \text{eff}(\text{move-right}_{t,i,j}) &= \{p_{i,j} \mapsto b, p_{i+1,j} \mapsto t\} \end{aligned}$$

Exercise 10.3 (1+0.5+0.5+1 marks)

The following graph depicts the state space of a cannibals and missionary problem with two cannibals and two missionaries:

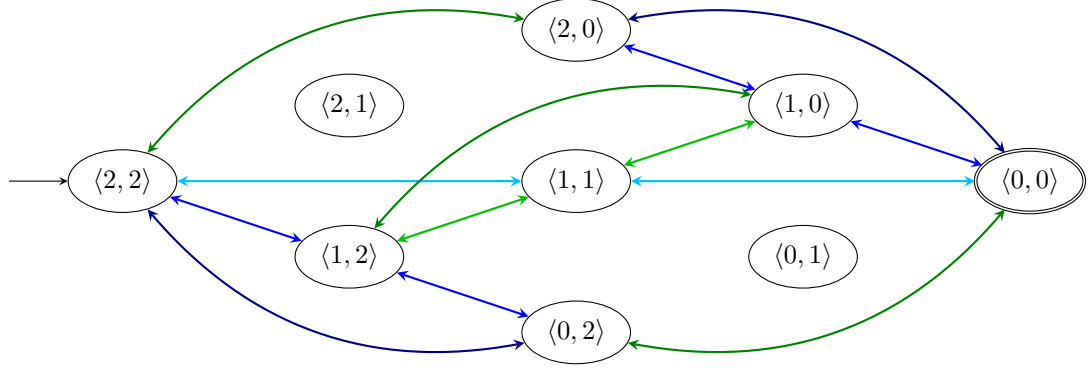


A state is represented as a tuple $\langle C, M, B \rangle$, where C and M (with $\text{dom}(C) = \text{dom}(M) = \{0, 1, 2\}$) denote how many cannibals and missionaries respectively are on the left shore, and B (with $\text{dom}(B) = \{L, R\}$) denotes on which shore the boat is. The cost of all operators is 1.

- (a) Draw the abstract state space that is induced by the projection to $P = \{C, M\}$ in the same way as the above depiction of the concrete state space. (In particular, draw a new graph containing only abstract states instead of indicating in the one above which states are mapped to the same abstract state.)

- (b) Use the abstraction from part (a) to derive a pattern database heuristic. Provide all database entries, i.e., the abstract distances for all states in the abstract state space.
- (c) What is $h^{\{C,M\}}(I)$?
- (d) For a cannibals and missionaries problem with n cannibals and n missionaries, what would $h^{\{C,M\}}(I)$ be? Justify your answer.

Solution:



(a)

(b) The PDB has the following entries:

$\langle 2, 2 \rangle \mapsto 2$	$\langle 2, 1 \rangle \mapsto \infty$	$\langle 1, 2 \rangle \mapsto 2$
$\langle 1, 1 \rangle \mapsto 1$	$\langle 2, 0 \rangle \mapsto 1$	$\langle 0, 2 \rangle \mapsto 1$
$\langle 1, 0 \rangle \mapsto 1$	$\langle 0, 1 \rangle \mapsto \infty$	$\langle 0, 0 \rangle \mapsto 0$

(c) $h^{\{C,M\}}(I) = 2$

(d) For problems with n cannibals and missionaries, we have $h^{\{C,M\}}(I) = n$. Since the position of the boat is ignored, we never need to return the boat from the right shore to the left. This enables us to apply “1 cannibal, 1 missionary” from the left to the right shore n times in succession, at which point all n cannibals and n missionaries are on the right shore. Since this action uses the full capacity of the boat, there can also not be any cheaper plan.